

EHR-RAGp: Retrieval-Augmented Prototype-Guided Foundation Model for Electronic Health Records

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Abstract

Electronic Health Records (EHR) contain rich longitudinal patient information and are widely used in predictive modeling applications. However, effectively leveraging historical data remains challenging due to long trajectories, heterogeneous events, temporal irregularity, and the varying relevance of past clinical context. Existing approaches often rely on fixed windows or uniform aggregation, which can obscure clinically important signals. In this work, we introduce EHR-RAGp, a retrieval-augmented foundation model that dynamically integrates the most relevant patient history across diverse clinical event types. We propose a prototype-guided retrieval module that acts as an alignment mechanism and estimates the relevance of retrieved historical chunks with respect to a given prediction task, guiding the model towards the most informative context. Across multiple clinical prediction tasks, EHR-RAGp consistently outperforms state-of-the-art EHR foundation model and transformer-based baselines. Furthermore, integrating EHR-RAGp with existing clinical foundation models yields substantial performance gains. Overall, EHR-RAGp provides a scalable and efficient framework for leveraging long-range clinical context to improve downstream performance.

1 Introduction

Healthcare systems rely extensively on Electronic Health Record (EHR) databases to store diverse clinical events collected during patient encounters, such as demographics, diagnoses, vital signs, lab results, procedures and medications [1, 2]. Recent advances in machine learning have enabled the adaptation of foundation models for EHR data, where patient trajectories are represented as chronologically ordered sequences of events, in a manner that mimics the structures of sentences in natural language [3, 4, 5, 6]. However, unlike natural language, clinical events are inherently irregular, sparse, fragmented, and span multiple episodes of care that accumulate over months or years [7].

Existing EHR foundation models are designed under specific assumptions, including: (i) information present only within a predefined time window for a specific patient encounter is relevant to predict the outcome of interest [8, 9]; and (ii) a preselected subset of events or features, such as lab test results, vital sign measurements, diagnoses, or combinations thereof, is sufficient to predict patient outcomes [10, 11, 12, 13, 14]. This often stems from computational constraints and the difficulty of representing long-range irregular patient trajectories. Such constraints could also lead to the omission of clinically important historical events. From a clinical perspective, the patient’s state is almost always shaped by their accumulated longitudinal history [15, 16]. This includes slow-moving chronic patterns and cross-encounter interactions that clinicians routinely examine when making decisions. Such a central gap highlights a fundamental limitation in current EHR modeling paradigms, namely the inability to integrate and reason over a patient’s full history within a unified representation learning framework.

Filling this gap requires a new outlook on EHR representation learning. Rather than limiting the patient’s predictive context to a predefined time window, a model must be equipped to access, retrieve,

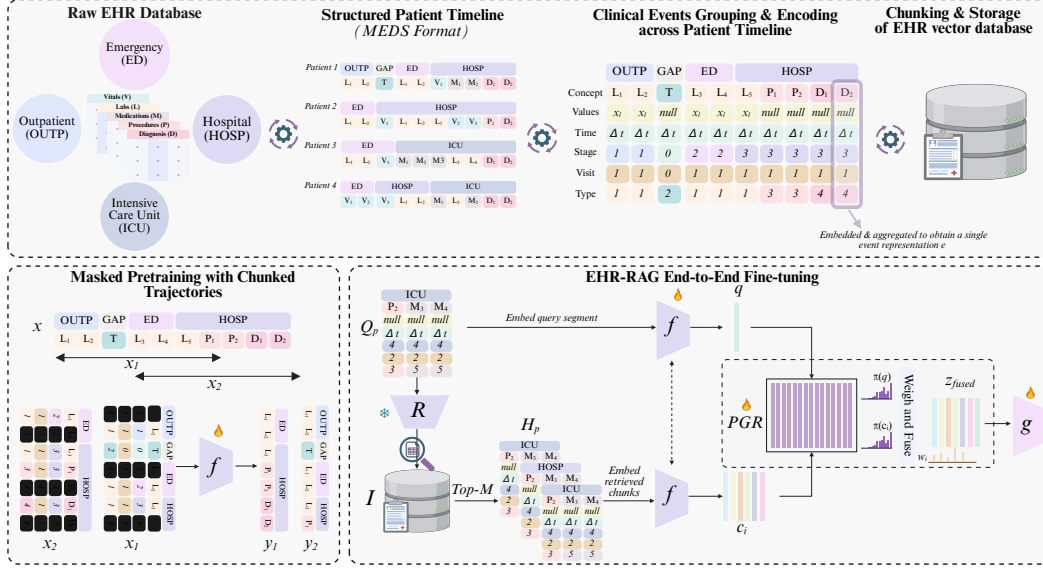


Figure 1: Overview of the EHR-RAGp, including chunking of patient trajectories to build the EHR vector database (*top panel*), masked pre-training of the backbone encoder with chunked trajectories (*bottom left*), and end-to-end fine-tuning of the final architecture (*bottom right*).

and integrate information scattered across the patient’s complete trajectory. The central assumption motivating this work is that retrieval provides a scalable mechanism for incorporating long-range historical context without forcing the model to encode the entire trajectory within its fixed internal state. Given the remarkable success of Retrieval-Augmented Generation [17] (RAG) techniques in Natural Language Processing (NLP) and their ability to leverage external knowledge beyond the training data of the base Large Language Model (LLM) [18, 19, 20, 21], the same concept can be applied for retrieving and integrating historical EHR data. In other words, by considering the patient’s historical events as retrievable pieces of information, an EHR foundation model can dynamically condition on what matters the most for the patient’s current state. This perspective shifts longitudinal modeling from a purely sequential problem into a retrieval-augmented one. Hence, in this work, we propose EHR-RAGp, a retrieval-based foundation model designed to reason over the full breadth of a patient’s longitudinal EHR history. The main contributions of the work are summarized as follows:

1. We propose a new framework for constructing a vector database of longitudinal EHR patient trajectories using multi-granular chunking strategies, including event-based, time-based, visit-level, and care-stage segmentation, enabling flexible representation of heterogeneous events, temporal structure, and care progression.
2. We introduce a prototype-guided retrieval mechanism that acts as an alignment operator, estimating the relevance of retrieved historical patient chunks with respect to the prediction task, guiding the model to integrate the most informative long-range clinical context.
3. We present EHR-RAGp, an EHR-native retrieval-augmented foundation model trained on large-scale real-world EHR data. Extensive evaluation across multiple clinical prediction tasks demonstrates that EHR-RAGp consistently outperforms multiple state-of-the-art models.
4. By training and optimizing all baselines using an expanded and unified feature set, we establish new benchmark results on four clinical prediction tasks and make our code publicly available at: <https://github.com/nyuad-cai/EHR-RAGp>.

2 Related Work

EHR Foundation Models. Traditional EHR modeling approaches process patient trajectories as sequences of discrete clinical events collected over time. Some models primarily consider the occurrence of certain events, while others process the occurrence of the clinical event and its measured value. Early techniques utilized recurrent models such as Long-Short Term Memory (LSTM) [22] or Gated Recurrent Unit (GRU) networks [23] to capture long-term dependencies [24, 25, 16, 26].

The emergence of transformer-based LLMs [27, 28] paved the way for more advanced EHR foundation models. For example, encoder-based models such as BEHRT [29], Hi-BEHT [30], CEHR-BERT [31], and Med-BERT [32] introduced large-scale pretraining over longitudinal sequences of medical events to enhance the contextualization of patient trajectories across visits. Other decoder-based transformer models such as ETHOS [33], CEHR-GPT [34], Curiosity [35], and EHR2Path [36] focused on the development of generative EHR foundation models for clinical prediction tasks. The common link between all these models is that they mimic standard language models by learning contextualized medical code vocabularies as tokens.

Another category of EHR foundation models shifts away from code-based modeling towards text-based modeling by treating each medical code as a textual description. Hence, the patient timeline constitutes a set of textual descriptions that can be processed directly by the LLM [12, 13, 37]. Despite the remarkable progress, these models continue to inherit core limitations from their language-model roots. They operate on truncated encounter windows, compress long and irregular patient histories into fixed-length sequences, and rely on restricted types of clinical events. While some methods have explored the use of longer context windows to accommodate extended trajectories [35, 12, 13], their vocabularies are heavily tokenized for medical codes and numerical values, or textualized and dependent on the base model’s reasoning capabilities. As a result, substantial portions of a patient’s longitudinal trajectory remain inaccessible in downstream prediction tasks. This emphasizes the need for scalable frameworks that are capable of integrating and reasoning over complete EHR trajectories.

Retrieval-Augmented Modeling. RAG has emerged as a powerful technique in NLP for extending large language models capabilities by allowing them to condition on external, non-parametric knowledge sources. The original RAG framework [17] demonstrated how combining dense retrieval with generative transformers improves factual correctness and reduces hallucinations. The concept was later extended by models such as RETRO [38], FLARE [39], and REPLUG [40]. These architectures share a common principle: retrieval allows the model to incorporate long-range or domain-specific information without encoding the entire knowledge base within its internal parameters. The success of the retrieval paradigm has also influenced healthcare, particularly in clinical question answering [41, 42], evidence grounding [43, 44], biomedical knowledge integration [45, 46], phenotyping and cohort identification [47], and many other applications related to medical textual data [48, 49].

In the context of EHR modeling, RAG has been primarily explored as a mechanism for integrating external knowledge into predictive models, such as EMERGE [50], RAM-EHR [51], REALM [52], and KAMELEON [53]. To the best of our knowledge, REMED [54] is our most closely related work. Its core idea is to retrieve the most relevant subset of medical codes for a given outcome prediction task, to overcome manual selection of clinical events or features. Overall, recent advancements highlight the potential of retrieval-augmented modeling for EHR, but they also reveal a clear unmet need for mechanisms that can dynamically access and integrate a patient’s entire structured history.

3 Methodology

3.1 Overview

EHR-RAGp is a retrieval-augmented foundation model for reasoning over longitudinal EHR trajectories (Figure 1). The framework encodes structured clinical events, organizes them into retrievable chunks, retrieves relevant historical segments through prototype-guided alignment, and integrates them with a task-specific query for downstream prediction. Here, we describe the core architectural components.

3.2 Preliminaries

Let $\mathcal{D} = \{\mathcal{H}_p\}_{p=1}^{\mathcal{P}}$ represent a large-scale EHR dataset consisting of patient-level longitudinal records for a cohort of $\mathcal{P}$ patients. For a given patient p , the longitudinal medical history is represented as:

$$\mathcal{H}_p = \{\mathcal{V}_p^{(1)}, \mathcal{V}_p^{(2)}, \dots, \mathcal{V}_p^{(N_p)}\},$$

where N_p denotes the total number of visits for the respective patient (see Figure S1). The n -th visit $\mathcal{V}_p^{(n)}$ consists of a set of clinical events:

$$\mathcal{V}_p^{(n)} = \{(c_{nk}, v_{nk}, t_{nk})\}_{k=1}^{K_n},$$

where c_{nk} denotes a clinical event such as a diagnosis, procedure, laboratory test, vital sign, or medication administration, v_{nk} is an optional associated value such as a lab result, medication dosage,

or vital sign measurement, and t_{nk} is the exact timestamp at which the event occurred. Events within a visit may vary in type, granularity, and frequency, reflecting the inherently irregular and heterogeneous nature of structured EHR data.

For a given index visit, denoted by $\mathcal{V}_p^{(\tau)}$, the goal is to predict a target patient outcome based on a true label $y_p^{(\tau)}$. Hence, the proposed system role is to use the index visit information in retrieving the most relevant information. Therefore, we define a **query segment** $\mathcal{Q}_p^{(\tau)} \subseteq \mathcal{V}_p^{(\tau)}$, which represents the portion of the current visit that is used by the model as its immediate input context. The query segment may correspond to only a portion of the full visit, as the entire visit can exceed the maximum input length supported by the model. Therefore, we restrict the query to a fixed-size segment that fits within the model’s context window. All events occurring before the query segment, either from the same visit or from any earlier visits, form the **historical context**:

$$\mathcal{H}_p^{(<\tau)} = \{\mathcal{V}_p^{(1)}, \dots, \mathcal{V}_p^{(\tau-1)}, \mathcal{V}_p^{(\tau < q)}\},$$

where $\mathcal{V}_p^{(\tau < q)}$ denotes the prefix of the last visit preceding the query (see Figure S2). Hence, the model aims to learn a mapping function:

$$\gamma : (\mathcal{Q}_p^{(\tau)}, \mathcal{H}_p^{(<\tau)}) \rightarrow \hat{y}_p^{(\tau)}.$$

A naive formulation of this problem would require the model to encode the full patient history in its fixed internal state, which is a significant challenge due to computational constraints. Hence, the goal of our proposed framework is to enable the model to retrieve and integrate the most relevant information from the patient’s history $\mathcal{H}_p^{(<\tau)}$ based on the query segment $\mathcal{Q}_p^{(\tau)}$.

3.3 Event Representation Learning

EHR data contains heterogeneous clinical events with varying semantics, numerical values, temporal patterns, and contextual metadata. To process these events within a unified architecture, we propose mapping each event (c_{nk}, v_{nk}, t_{nk}) and its associated metadata into a single representation $e_{nk} \in \mathbb{R}^d$, where d is the vector dimension. This embedding considers six key components: clinical concept $\mathbf{u}_{c_{nk}}$, numeric value $\mathbf{v}_{nk}$, time $\mathbf{t}_{nk}$, visit-order $\mathbf{r}_{\text{visit}(n)}$, care stage $\mathbf{s}_{\text{stage}(n)}$, and event-type $\mathbf{w}_{\text{type}(nk)}$ embeddings as shown below. Detailed information pertaining to data encoding is provided in Appendix B. The unified event representation is then constructed by summation following [10, 29]:

$$e_{nk} = (\mathbf{u}_{c_{nk}} + \mathbf{v}_{nk} + \mathbf{t}_{nk} + \mathbf{r}_{\text{visit}(n)} + \mathbf{s}_{\text{stage}(n)} + \mathbf{w}_{\text{type}(nk)}) \in \mathbb{R}^d$$

This unified representation enables encoding of heterogeneous events, longitudinal ordering, and care-setting context within a single embedding space.

3.4 Chunking

After representing individual events, EHR-RAGp partitions the patient timeline into coherent chunks that serve as retrievable units. We propose four chunking methods, namely, *event-based*, *time-based*, *visit-level*, and *care-stage* chunking (Figure S3). Each chunking method captures a different aspect of longitudinal patient structure, enabling retrieval over heterogeneous temporal granularities and clinical contexts. Event-based chunking groups a fixed number of consecutive events, time-based chunking partitions the timeline into fixed temporal windows, visit-level chunking organizes events according to hospital admissions, and care-stage chunking separates events based on transitions between clinical care settings, such as outpatient, emergency department, hospital ward, and ICU.

3.5 Prototype-Guided Retrieval

Given a query segment $\mathcal{Q}_p^{(\tau)}$, EHR-RAGp retrieves relevant historical segments from the patient’s longitudinal history $\mathcal{H}_p^{(<\tau)}$ using a two-stage procedure: (1) coarse-grained semantic retrieval to identify candidate chunks, followed by (2) prototype-guided alignment that estimates the relevance of each retrieved chunk via agreement in the learned prototype assignment space, to enable weighted integration of context for the downstream task.

Candidate Retrieval. The query segment $Q_p^{(\tau)}$ is first encoded using a frozen pretrained retriever R , producing a fixed-dimensional query representation:

$$q' = R(Q_p^{(\tau)}) \in \mathbb{R}^d.$$

The retriever, R , is pretrained using masked language modeling. Further details are provided in Appendix C. Using the same retriever R , each historical chunk $h_i \in \mathcal{H}_p^{(<\tau)}$ is embedded as

$$c'_i = R(h_i) \in \mathbb{R}^d.$$

Candidate retrieval is then performed by computing **cosine** similarity between the query embedding q' and historical chunk embeddings $\{c'_i\}$ stored in precomputed similarity index $\mathcal{I}$. We then retrieve the top- M most similar chunks:

$$\{h_1, h_2, \dots, h_M\} \subset \mathcal{H}_p^{(<\tau)}.$$

Query and Candidate Encoding. Upon retrieval, the query segment $Q_p^{(\tau)}$ and the top- M chunks $\{h_1, h_2, \dots, h_M\}$ are passed to a shared backbone encoder f to obtain q and $\{c_i\}$, respectively.

$$q = f(Q_p^{(\tau)}) \in \mathbb{R}^d, \quad c_i = f(h_i) \in \mathbb{R}^d.$$

Using the same encoder f for queries and historical chunks ensures that both are represented in a common embedding space, enabling consistent comparison and alignment between the query and retrieved context.

Prototype-Guided Alignment. EHR-RAGp maintains a learnable set of L prototype vectors, which serve as latent clinical centroids that organize a patient’s longitudinal history into distinct semantic modes:

$$P = \{p_1, p_2, \dots, p_L\}, \quad p_l \in \mathbb{R}^d,$$

These prototypes serve as latent anchors that organize the representation space into distinct modes, enabling structured comparison between query and historical segments. Rather than being predefined, they are learned end-to-end, allowing the model to capture recurring patterns in clinical trajectories and facilitate alignment between the query and retrieved context with respect to the downstream task.

For a chunk embedding $x \in \mathbb{R}^d$, the model computes a probability distribution over the L prototypes:

$$\pi(x) = \text{softmax}\left(\frac{xP^\top}{T_\pi}\right) \in \mathbb{R}^L,$$

where x corresponds to either to the **query** q or a **candidate chunk** c_i , and T_π is a temperature parameter that controls the sharpness of the distribution [55]. In practice, we use different temperatures for the query and history branches, denoted by T_q and T_h , respectively, with $T_q < T_h$. This asymmetric design reflects the roles of the two branches: the query must identify a focused subset of relevant patterns for the prediction task, while historical chunks represent diverse clinical contexts that may only partially match the query. Accordingly, we obtain

$$\pi_q = \pi(q), \quad \pi_i = \pi(c_i), \quad i = 1, \dots, M.$$

π_q represents the prototype assignment distribution of the query, while each π_i represents the assignment of candidate chunk c_i to the set of learned prototypes L . To measure how well a candidate history chunk aligns with the query in the prototype space, EHR-RAGp measures the agreement between their prototype distributions using a cross-entropy-based alignment score:

$$\alpha_i = - \sum_{l=1}^L \pi_q^{(l)} \log \pi_i^{(l)},$$

Where $\alpha \in [0, \infty)$. This formulation measures how well the candidate distribution π_i explains the query distribution π_q , treating the query as a reference. Lower negative cross-entropy values indicate stronger alignment, meaning that the candidate assigns high probability mass to the prototypes emphasized by the query, whereas higher values indicate weaker alignment. In addition, this distributional alignment enables smooth and differentiable matching in the prototype space without requiring hard assignment to a single prototype.

Prototype-Guided Weighting. Given the prototype alignment scores $\{\alpha_i\}_{i=1}^M$, EHR-RAGp converts them into normalized importance weights using a temperature-controlled softmax:

$$w_i = \text{softmax}\left(\frac{-\alpha_i}{T_s}\right),$$

where T_s is a temperature parameter that controls the smoothness of the weighting distribution.

EHR-RAGp adopts a fully differentiable soft weighting mechanism, where all retrieved chunks contribute to the final prediction with varying degrees of importance. This design allows the model to preserve distributed clinical evidence across multiple chunks, which is particularly important in EHR data where relevant signals are often weak and spread over time. The resulting weights $\{w_i\}$ are used to modulate the contribution of each retrieved chunk during the subsequent fusion stage

Retrieval-Augmented Fusion. The query representation q and the retrieved historical chunks $\mathcal{R}_p^{(<\tau)} = \{c_1, \dots, c_n\}$ are combined through a prototype-guided weighting mechanism. Each retrieved chunk is modulated by its corresponding importance weight w_i :

$$z_{\text{fused}} = [q, w_1 c_1, w_2 c_2, \dots, w_n c_n].$$

This sequence is then processed by a transformer encoder $g(\cdot)$ followed by a linear classifier $h(\cdot)$ to yield the final prediction:

$$\hat{y}_p^{(\tau)} = h(g(z_{\text{fused}})).$$

Training Objective. EHR-RAGp is trained end-to-end using a supervised binary cross-entropy loss associated with the target prediction task, while keeping the retrieval encoder, R , frozen and updating the backbone encoder, f , the prototype parameters, and the fusion module. The overall objective combines the task loss with a prototype usage regularization term that encourages balanced utilization of the prototype space across both query and history representations:

$$\mathcal{L} = l(y_p^{(\tau)}, \hat{y}_p^{(\tau)}) - \lambda_u (\mathcal{H}(\bar{\pi}_q) + \mathcal{H}(\bar{\pi}_h)),$$

where $\bar{\pi}_q = \frac{1}{B} \sum_{b=1}^B \pi_q^{(b)}$ and $\bar{\pi}_h = \frac{1}{\sum_b K_b} \sum_{b,k} \pi_{h_k}^{(b)}$ denote the average prototype assignment distributions over a batch for the query and historical chunks, respectively, $\mathcal{H}(\cdot)$ is the entropy function, and $\lambda_u > 0$ determines the strength of the regularization term. This regularization encourages the model to avoid degenerate solutions in which only a small subset of prototypes are used.

4 Experiments

Dataset & Prediction Tasks. We evaluate our proposed framework on multiple tasks using the large-scale real-world MIMIC-IV dataset (V3.1) [56]. MIMIC-IV consists of heterogeneous EHR data gathered from patients admitted to the Beth Israel Deaconess Medical Center between 2008-2022. It includes clinical data collected from 364,627 patients associated with 546,028 hospital admissions and 94,458 ICU stays [57]. It consists of two main modules: *hosp* and *icu*. While the former covers general admission data such as labs, medications, and microbiology events, the latter covers ICU-level charted data such as chartevents, infusions, and fluid outputs. We consider all tables that hold clinically relevant information and exclude those tables holding operational information such as provider information. Table 1 summarizes the data statistics with further information provided in Appendix B.

We assess EHR-RAGp for four risk prediction tasks: (1) in-hospital mortality, (2) ICU-readmission within 30 days, (3) long length of stay (7 days or more), and (4) 1-year mortality post-discharge. The selected tasks span both short-term and long-term clinical outcomes and reflect common benchmarks in EHR-based modeling. Together, they assess the model’s ability to capture acute risk, longitudinal disease progression, and healthcare utilization by leveraging structured patient histories. Detailed information pertaining to the tasks and data preprocessing are provided in Appendix C.2.

Table 1: Summary of data splits used for downstream evaluation.

Statistic	Training	Val	Test	Total
# Patients	34, 887	4, 984	9, 968	49, 839
# ICU stays	42, 679	6, 133	12, 363	61,175
# ICU events	231M	33M	67M	332M
Mean LOS (days)	4.164	4.160	4.170	—
Percentage (%)	70	10	20	100

Table 2: Performance comparison between EHR-RAGp and all baselines. Best results are marked in **bold**. Second best results are underlined.

Model	ICU-Readmit 30d		In-hospital Mortality		Long LOS 7d		1YR Mortality	
	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)
Clinical baselines								
DescEmb [58]	0.660 (0.634, 0.684)	0.096 (0.080, 0.118)	0.933 (0.927, 0.940)	0.693 (0.669, 0.716)	0.849 (0.840, 0.858)	0.480 (0.455, 0.505)	0.766 (0.755, 0.777)	0.329 (0.308, 0.351)
DescEmb [†] [58]	0.632 (0.605, 0.658)	0.098 (0.080, 0.124)	0.911 (0.903, 0.918)	0.631 (0.606, 0.655)	0.815 (0.804, 0.825)	0.403 (0.378, 0.427)	0.740 (0.728, 0.752)	0.304 (0.283, 0.327)
GenHPF [13]	0.705 (0.681, 0.728)	0.134 (0.108, 0.167)	0.932 (0.926, 0.939)	0.706 (0.682, 0.730)	0.850 (0.841, 0.859)	0.482 (0.456, 0.510)	0.781 (0.770, 0.793)	0.345 (0.324, 0.369)
Med-BERT [32]	0.715 (0.693, 0.739)	0.106 (0.091, 0.125)	0.907 (0.899, 0.915)	0.609 (0.581, 0.635)	0.802 (0.790, 0.814)	0.387 (0.364, 0.411)	0.749 (0.738, 0.760)	0.305 (0.285, 0.327)
CEHR-BERT [31]	0.718 (0.694, 0.739)	0.133 (0.110, 0.163)	0.933 (0.927, 0.940)	0.700 (0.677, 0.724)	0.838 (0.829, 0.848)	0.447 (0.423, 0.472)	0.778 (0.767, 0.789)	0.342 (0.321, 0.365)
BEHRT [29]	0.737 (0.714, 0.761)	0.133 (0.112, 0.162)	0.925 (0.918, 0.932)	0.674 (0.648, 0.697)	0.831 (0.821, 0.841)	0.439 (0.417, 0.463)	0.771 (0.760, 0.783)	0.337 (0.315, 0.361)
Hi-BEHRT [30]	0.596 (0.571, 0.621)	0.055 (0.048, 0.067)	0.897 (0.888, 0.904)	0.585 (0.557, 0.611)	0.775 (0.761, 0.787)	0.355 (0.332, 0.380)	0.700 (0.688, 0.712)	0.254 (0.237, 0.273)
Long context baselines								
EHRMamba [10]	0.725 (0.702, 0.749)	0.130 (0.109, 0.160)	0.925 (0.918, 0.932)	0.640 (0.614, 0.670)	0.865 (0.855, 0.874)	0.578 (0.552, 0.605)	0.757 (0.746, 0.769)	0.303 (0.283, 0.325)
ModernBERT [59]	0.740 (0.715, 0.762)	0.137 (0.114, 0.167)	0.937 (0.931, 0.943)	0.713 (0.688, 0.735)	0.869 (0.86, 0.879)	0.597 (0.573, 0.624)	0.807 (0.797, 0.817)	0.370 (0.349, 0.395)
Transformer-based baselines								
RoBERTa [60]	0.734 (0.710, 0.756)	0.141 (0.118, 0.173)	0.925 (0.917, 0.932)	0.671 (0.644, 0.697)	0.842 (0.833, 0.851)	0.476 (0.453, 0.501)	0.781 (0.770, 0.791)	0.345 (0.325, 0.369)
LongFormer [61]	0.736 (0.713, 0.760)	0.123 (0.105, 0.153)	0.922 (0.914, 0.929)	0.645 (0.615, 0.673)	0.855 (0.846, 0.864)	0.546 (0.521, 0.570)	0.789 (0.778, 0.800)	0.356 (0.336, 0.382)
BigBird [62]	0.724 (0.700, 0.747)	0.133 (0.109, 0.164)	0.931 (0.925, 0.938)	0.690 (0.667, 0.714)	0.864 (0.855, 0.874)	0.559 (0.534, 0.585)	0.787 (0.777, 0.798)	0.355 (0.332, 0.377)
RoFormer [63]	0.739 (0.716, 0.760)	0.142 (0.116, 0.173)	0.935 (0.928, 0.942)	0.701 (0.679, 0.724)	0.872 (0.863, 0.880)	0.568 (0.542, 0.593)	0.798 (0.788, 0.808)	0.364 (0.342, 0.387)
Retrieval-based models								
REMed [54]	0.535 (0.510, 0.561)	0.044 (0.039, 0.052)	0.867 (0.857, 0.876)	0.468 (0.439, 0.499)	0.800 (0.790, 0.811)	0.363 (0.342, 0.386)	0.622 (0.608, 0.636)	0.185 (0.175, 0.199)
Vanilla EHR-RAGp	0.742 (0.719, 0.764)	0.146 (0.121, 0.177)	0.939 (0.933, 0.945)	0.710 (0.687, 0.732)	0.879 (0.868, 0.888)	0.618 (0.592, 0.642)	0.814 (0.804, 0.824)	0.381 (0.359, 0.405)
EHR-RAGp (Ours)	0.747 (0.724, 0.768)	0.156 (0.128, 0.189)	0.940 (0.933, 0.945)	0.716 (0.693, 0.738)	0.885 (0.876, 0.893)	0.628 (0.603, 0.652)	0.821 (0.811, 0.831)	0.396 (0.372, 0.422)

[†] DescEmb represents DescEmb BERT-FT variant, whereas DescEmb* represents DescEmb CLS-FT variant. For more information, see Appendix D.

Table 3: Performance gains of existing EHR baseline with EHR-RAGp.

Model	EHR-RAGp	ICU-Readmit 30d		In-hospital Mortality		Long LOS 7d		1YR Mortality	
		AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)
Med-BERT	×	0.715 (0.693, 0.739)	0.106 (0.091, 0.125)	0.907 (0.899, 0.915)	0.609 (0.581, 0.635)	0.802 (0.790, 0.814)	0.387 (0.364, 0.411)	0.749 (0.738, 0.760)	0.305 (0.285, 0.327)
	✓	0.725 (0.703, 0.748)	0.121 (0.103, 0.147)	0.929 (0.922, 0.936)	0.683 (0.658, 0.709)	0.861 (0.851, 0.870)	0.586 (0.561, 0.610)	0.773 (0.761, 0.784)	0.307 (0.289, 0.328)
CEHR-BERT	×	0.718 (0.694, 0.739)	0.133 (0.110, 0.163)	0.933 (0.927, 0.940)	0.700 (0.677, 0.724)	0.838 (0.829, 0.848)	0.447 (0.423, 0.472)	0.778 (0.767, 0.789)	0.342 (0.321, 0.365)
	✓	0.745 (0.724, 0.767)	0.167 (0.137, 0.203)	0.939 (0.933, 0.946)	0.717 (0.694, 0.741)	0.876 (0.867, 0.885)	0.604 (0.580, 0.627)	0.802 (0.791, 0.811)	0.372 (0.350, 0.396)
BEHRT	×	0.737 (0.714, 0.761)	0.133 (0.112, 0.162)	0.925 (0.918, 0.932)	0.674 (0.648, 0.697)	0.831 (0.821, 0.841)	0.439 (0.417, 0.463)	0.771 (0.760, 0.783)	0.337 (0.315, 0.361)
	✓	0.743 (0.722, 0.765)	0.135 (0.112, 0.166)	0.932 (0.925, 0.939)	0.680 (0.656, 0.704)	0.871 (0.862, 0.879)	0.599 (0.575, 0.623)	0.792 (0.783, 0.803)	0.361 (0.338, 0.385)

Implementation Details. We use RoFormer *base* [63] as the backbone encoder for EHR-RAGp. In the first stage, we pretrain the encoder using an MLM objective [60]. For fairness of comparison, all code-based baselines in our evaluation are pretrained using the same objective and data splits. The pretrained EHR-RAGp encoder is subsequently reused as the embedding model for constructing and querying the vector database. In the second stage, we fine-tune the full EHR-RAGp model end-to-end for the downstream prediction tasks in a supervised setting. We conduct extensive hyperparameter tuning using Bayesian optimization for all baselines and EHR-RAGp models. Detailed optimization settings and training schedules are provided in Appendix C. We evaluate all models using the Area Under the Operator Receiver Curve (AUROC) and the Area Under the Precision-Recall (AUPRC) and report them along with %95 confidence interval (CI) using bootstrapping [64].

Baselines. We consider a diverse set of baselines spanning clinical EHR models, general transformer architectures, long-context models, retrieval-based methods, and LLMs. Clinical baselines include existing EHR-specific foundation models such as DescEmb [58], GenHPF [13], Med-BERT [32], CEHR-BERT [31], BEHRT [29], and Hi-BEHRT [30]. Since these models were originally trained on different subsets of clinical features, we adapt all baselines using a shared vocabulary and EHR embedding layer to ensure unified input representations and fair comparison. We further include general transformer-based encoders adapted to EHR data, including RoBERTa [60], Longformer [61], BigBird [62], and RoFormer [63]. For long-context modeling, we consider EHRMamba [10], designed for sequences up to 2048 events, and ModernBERT [59], supporting contexts up to 8192 tokens. Retrieval baselines include REMed [54] and a vanilla RAG variant without prototype-guided refinement. Finally, we evaluate both general-purpose (Qwen2.5-7B [65], Mistral-7B [66]) and medically adapted LLMs (MedGemma-1.5-4B [67], BioMistral-7B [68]) in a zero-shot setting. Additional details are provided in Appendix D.

5 Main Results

Overall Performance Results. Table 2 reports the downstream performance across four clinical prediction tasks. EHR-RAGp consistently achieves the best results across all metrics, outperforming strong baselines from multiple model categories. For ICU readmission, EHR-RAGp attains an AUROC of 0.747 and AUPRC of 0.156, improving over the strongest transformer baseline RoFormer and

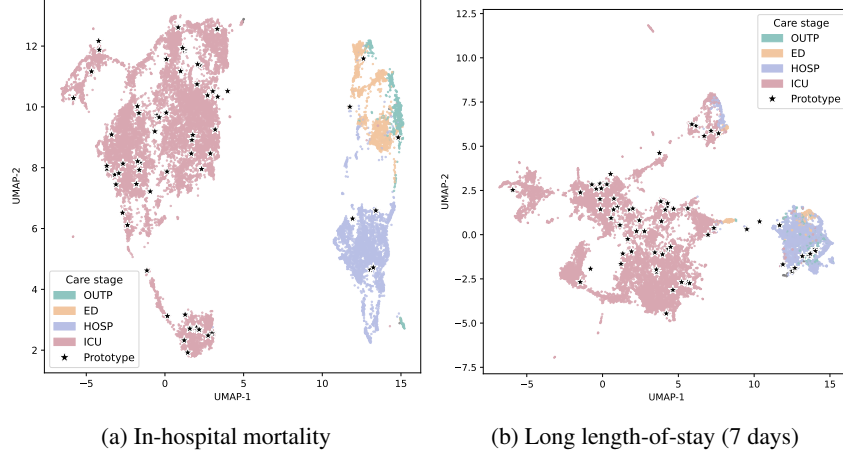


Figure 2: UMAP projections of history embeddings and prototypes for (a) long length of stay and (b) in-hospital mortality tasks.

long-context model ModernBERT. For in-hospital mortality, EHR-RAGp achieves the best performance (AUROC 0.940, AUPRC 0.716), surpassing ModernBERT (AUPRC 0.713) and CEHR-BERT (AUPRC 0.700). For long length-of-stay, EHR-RAGp reaches AUROC 0.885 and AUPRC 0.628, outperforming both long-context baselines, such as EHRMamba (AUROC 0.865, AUPRC 0.578), and transformer models like RoFormer (AUROC 0.872, AUPRC 0.568). Similarly, for 1-year mortality, it achieves an AUROC of 0.821 and AUPRC 0.396, exceeding RoFormer (AUROC 0.798, AUPRC 0.364) and ModernBERT (AUROC 0.807, AUPRC 0.370). Compared to clinical foundation models, the gains are substantial across all tasks. While the vanilla version without prototypes consistently underperforms the full model, it still outperforms other baselines almost on all tasks. Overall, the results demonstrate that EHR-RAGp provides consistent improvements over all baselines, indicating effective utilization of long-range clinical context beyond increasing context length alone. Table S5 shows that LLM baselines perform substantially worse across all tasks.

EHR-RAGp Improves Clinical Foundation Models. Table 3 evaluates whether EHR-RAGp can improve existing EHR foundation models when used as a retrieval-augmented extension. Across Med-BERT, CEHR-BERT, and BEHRT, adding EHR-RAGp consistently improves performance across all four tasks. For Med-BERT, the largest gains appear in long LOS prediction, where AUPRC increases from 0.387 to 0.586, and in-hospital mortality, where AUPRC improves from 0.609 to 0.683. For CEHR-BERT, EHR-RAGp improves ICU readmission AUPRC from 0.133 to 0.167, long LOS AUPRC from 0.447 to 0.604, and 1-year mortality AUPRC from 0.342 to 0.372. Similar gains are observed for BEHRT, especially for long LOS, where AUPRC increases from 0.439 to 0.599. These results show that EHR-RAGp is not only a standalone model, but also a general retrieval-augmentation framework that can strengthen existing EHR foundation models.

Latent Representation Structure. Figure 2 presents UMAP projections of history embeddings for two tasks: in-hospital mortality and long length-of-stay. Across both tasks, the embedding space exhibits a consistent global structure, with clear separation between high-acuity (ICU) and lower-acuity (HOSP, ED, OUP) regions, indicating that the model organizes patient history primarily along clinical severity. Additionally, task-specific variations are evident in the internal geometry of clusters, suggesting that the learned representations adapt to the prediction objective. Notably, prototype centers (black stars) are distributed across dense regions of the manifold, covering both dominant ICU clusters and smaller peripheral structures, which supports their role in capturing diverse modes of patient trajectories. Overall, the visualization highlights that the model learns a structured and task-sensitive representation space while preserving clinically meaningful organization across care stages. See Figure S4 for UMAP plots of ICU-Readmit 30d and 1YR Mortality tasks. To further analyze retrieval behavior and prediction dynamics, we provide qualitative analysis in Appendix E.3, illustrating prototype usage, chunk relevance weighting, and care-stage contributions at inference.

Table 5: Impact of choice of chunking strategy on model performance across all tasks.

Model	ICU-Readmit 30d		In-hospital Mortality		Long LOS 7d		1YR Mortality	
	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)
Event-Based	0.747 (0.724, 0.768)	0.156 (0.128, 0.189)	0.939 (0.933, 0.945)	0.710 (0.684, 0.732)	0.884 (0.875, 0.891)	0.611 (0.586, 0.634)	0.817 (0.807, 0.827)	0.390 (0.365, 0.416)
Time-Based	0.745 (0.722, 0.766)	0.152 (0.126, 0.185)	0.937 (0.930, 0.943)	0.706 (0.683, 0.730)	0.884 (0.876, 0.893)	0.623 (0.598, 0.647)	0.807 (0.797, 0.817)	0.377 (0.353, 0.402)
Visit-Level	0.738 (0.715, 0.762)	0.153 (0.125, 0.184)	0.939 (0.933, 0.945)	0.715 (0.692, 0.737)	0.885 (0.876, 0.893)	0.628 (0.603, 0.652)	0.821 (0.811, 0.831)	0.396 (0.372, 0.422)
Care-Stage	0.743 (0.720, 0.764)	0.150 (0.124, 0.182)	0.940 (0.933, 0.945)	0.716 (0.693, 0.738)	0.884 (0.875, 0.893)	0.618 (0.590, 0.645)	0.820 (0.811, 0.829)	0.393 (0.369, 0.418)

6 Ablations

Effect of Regularization. Table 4 evaluates the impact of prototype regularization (λ_u) on both representation behavior and downstream performance. Without regularization, prototype assignments are highly peaked for queries (e.g., $P_{\max} = 0.887$ for Long LOS), indicating collapse to a few dominant prototypes, while history assignments remain sparse and less informative. Enabling regularization significantly increases entropy and reduces $P_{\max}$ for both query and history, leading to more distributed and balanced prototype usage. This results in consistent performance gains across tasks, with Long LOS AUPRC improving from 0.579 to 0.618, and ICU readmission AUPRC from 0.118 to 0.156. Overall, the results show that regularization prevents prototype collapse, promotes better utilization of the prototype space, and improves performance. See Appendix E.4 for further analysis on the usage regularization λ_u .

Effect of Chunking Strategies. Table 5 presents an ablation over chunking strategies. Overall, performance is consistent across all strategies, indicating that EHR-RAGp is robust to how patient history is segmented. Event-based chunking achieves the best results for ICU readmission (AUROC 0.747, AUPRC 0.156), while care-stage chunking performs best for in-hospital mortality (AUROC 0.940, AUPRC 0.716). Visit-level chunking yields the strongest performance for long-term outcomes, including long LOS (AUPRC 0.628) and 1-year mortality (AUPRC 0.396). Time-based chunking performs competitively across all tasks but does not outperform other strategies. These results suggest that no single chunking scheme dominates universally, and different chunking methods capture complementary aspects of patient trajectories depending on the prediction task.

Effect of Prototypes Count. Table 6 shows the effect of varying the number of prototypes. Increasing the number of prototypes consistently improves performance for ICU readmission, with AUROC rising from 0.728 to 0.743, with 64 and 512 prototypes, and AUPRC from 0.132 to 0.150, indicating that a richer prototype set better captures fine-grained structure in patient trajectories. In contrast, performance on in-hospital mortality remains relatively stable across configurations, with marginal variations around AUROC and AUPRC. Overall, this suggests that the effective number of prototypes is also task dependent.

7 Discussion & Conclusion

In this work, we introduce EHR-RAGp, a retrieval-augmented, prototype-guided foundation model designed to reason over heterogeneous EHR trajectories by selectively incorporating relevant patient history. By framing longitudinal EHR modeling as a retrieval problem, EHR-RAGp addresses the limitations of fixed-window approaches and enables scalable integration of long-range clinical context. Across multiple downstream tasks, our approach consistently outperforms strong baselines as well as boosting existing clinical foundation models when used as an extension. Beyond quantitative improvements, our qualitative analyses further show that EHR-RAGp learns structured latent representations and dynamically reweighs retrieved patient history according to downstream relevance rather than retrieval similarity alone. Together, these results support the central claim of this work: increasing the effective medical context available to prediction models improves performance [6, 10, 30].

Limitations. While EHR-RAGp shows strong potential, some limitations remain to be investigated in future research. First, experiments are conducted on a single EHR dataset, and evaluation on different

Table 4: Ablation of regularization term with different number of prototypes (#p).

#p	λ_u	Query		History		Performance	
		P max	Entropy	P Max	Entropy	AUROC	AUPRC
Long LOS 7d							
64	×	0.887	0.663	0.094	4.048	0.866 (0.857, 0.876)	0.579 (0.552, 0.605)
	✓	0.084	4.096	0.027	4.156	0.884 (0.875, 0.893)	0.618 (0.590, 0.645)
ICU-Readmit 30d							
512	×	0.716	1.578	0.030	6.099	0.730 (0.709, 0.752)	0.118 (0.098, 0.144)
	✓	0.014	6.198	0.004	6.232	0.747 (0.724, 0.768)	0.156 (0.128, 0.189)

Table 6: Ablation of number of prototypes.

#p	ICU-Readmit 30d		In-hospital Mortality	
	AUROC	AUPRC	AUROC	AUPRC
64	0.728 (0.705, 0.751)	0.132 (0.109, 0.161)	0.937 (0.931, 0.943)	0.704 (0.680, 0.726)
128	0.736 (0.711, 0.757)	0.144 (0.121, 0.176)	0.937 (0.930, 0.943)	0.706 (0.683, 0.730)
256	0.739 (0.715, 0.761)	0.145 (0.121, 0.179)	0.934 (0.928, 0.94)	0.695 (0.672, 0.721)
512	0.743 (0.720, 0.764)	0.150 (0.124, 0.182)	0.935 (0.928, 0.941)	0.697 (0.673, 0.723)

datasets is needed to assess generalizability across different patient populations and coding systems. Second, experiments were conducted under a limited set of retrieval, and chunking configurations (e.g., chunk size of 256, 6-hour temporal windows, and Top-M of 24), further investigation of larger contexts and alternative chunking granularities is required. Third, although prototype-guided alignment improves performance and organizes the representation space into latent modes, the clinical interpretability of individual prototypes remains limited and requires deeper analysis with expert validation. Finally, while EHR-RAGp is designed on purpose for event-based EHR data, incorporation of other clinical modalities may provide complementary context for retrieval and prediction.

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A Broader Impact

Our work presents a step forward toward retrieval-augmented foundation models for structured EHR data. By enabling reasoning over a patient’s full longitudinal history, our approach may improve the performance of clinical prediction models and decision-support systems, with downstream applications in patient risk stratification, resource allocation, and personalized care. Such advances could support clinicians in managing complex medical histories and contribute to improved health-care outcomes. At the same time, models trained on sensitive health data raise important ethical considerations, including risks related to privacy, data bias, and inappropriate reliance on automated predictions. While our experiments use de-identified data and are intended for research purposes, real-world deployment would require careful validation, governance, and safeguards to ensure fairness, transparency, and patient safety. We encourage continued interdisciplinary collaboration to responsibly guide the development and use of retrieval-augmented EHR foundation models.

B Data Preprocessing

B.1 Cohort construction

Pretraining Cohort: We first convert the MIMIC-IV dataset into the Medical Event Data Standard (MEDS) format [69, 70], representing each patient history as a chronologically ordered sequence of medical events covering all patient visits within a single sequence. We then filter the dataset to construct the pretraining cohort. Specifically, we exclude patients listed in the patients table who do not have any valid hospital or ICU admissions. Next, we examine the total number of events within each visit and exclude visits containing fewer than 10 events. At this stage, we do not apply any age-based filtering, as this cohort is intended solely for pretraining. We further exclude all patients included in the test set from the pretraining cohort to prevent data leakage during evaluation. The resulting cohort consists of 199,012 patients with 463,436 hospital admissions, 80,005 ICU stays, and 496,199,673 medical events.

Downstream Cohort: For downstream tasks, we construct the dataset using ICU stays only and exclude patients with hospital admissions but no ICU stays. We follow the inclusion and exclusion criteria proposed by [25] to define the downstream cohort. First, we include all adult patients with an age at admission of ≥ 18 years. Second, we retain patients with a single ICU stay within the same hospital admission. Third, we include only patients whose admission and discharge ICU units are identical. Finally, we examine the length of each valid ICU stay and retain only those stays lasting at least 24 hours. This process results in a downstream cohort comprising 49,839 patients with 61,175 ICU stays.

B.2 Data Preprocessing

We perform comprehensive preprocessing to reduce irregularities and transform the data into a standardized format. First, MIMIC-IV contains ICD diagnosis and procedure codes from both versions 9 and 10. To unify all codes under a single coding system, we map all ICD-9 diagnosis and procedure codes to ICD-10. We use the General Equivalence Mappings (GEMS¹) released in 2018, which is the most recent version available.

Second, we examine medication events and remove any medications appearing in the patient timeline with ambiguous or unclear names. For laboratory tests, we first identify all lab entries listed in the MIMIC-IV file *d_labitems_to_loinc.csv* that are marked as non-laboratory measurements and filter them out. Subsequently, we modify lab event naming to incorporate the lab *item_id*, specimen fluid, and test name. This naming convention preserves the original structure of the MIMIC-IV dataset, increases the expressiveness of laboratory events, and reduces semantic ambiguity.

For ICU data, we modify event names to explicitly reflect the source table. For example, events originating from the *chartevents.csv* table are prefixed with *ICU-CHART*. We further include the corresponding *item_id* as a secondary identifier, followed by the event-specific name. Representative examples are provided in Table S1.

¹<https://www.cms.gov/medicare/coding-billing/icd-10-codes/icd-10-cm-icd-10-pcs-gem-archive>

Next, we handle outliers in numeric values using the *MEDS-transform* Python package by computing dataset-wide statistics for each numerical event and excluding values that lie beyond three standard deviations from the mean. Finally, we normalize all remaining numeric values using *MEDS-transform* by recomputing normalization statistics after outlier removal.

B.3 Patient Timeline Construction

Figure S1 depicts a sample patient timeline segmented into distinct care stages and covering all relevant clinical information. To construct a patient timeline, we first examine each patient sequence and identify hospital admissions (HOSP) using the *hadm_id* field, as well as ICU stays (ICU) using the *icustay_id* field, which are the only care stages in MIMIC-IV associated with unique identifiers.

During this process, we observe that certain medical events, such as laboratory tests and microbiology orders, appear in the patient timeline without an explicit reference to any hospital admission. To handle these events without excluding them, we assess their temporal proximity to admissions present in the patient timeline. Based on the time difference, we associate each such event with the nearest hospital admission by assigning the corresponding *hadm_id*. Events occurring within 24 hours of the nearest hospital admission are labeled as emergency department (ED) events, while events occurring within 30 days are treated as outpatient (OUTP) events.

To represent time gaps between consecutive visits, we insert artificial time tokens following prior work [31, 10], denoted as GAP, as illustrated in Figure S1. These tokens encode temporal intervals spanning weeks, months, or years, depending on the elapsed time between visits.

A standard patient timeline primarily consists of hospital events including laboratory measurements, medications, microbiology tests, ICD procedure codes, ICD diagnosis codes, and DRG codes, as well as ICU events such as chart events, infusions, procedures, and fluid outputs. In addition, administrative events are included for both hospital and ICU stays to indicate boundaries, locations, and care types. To explicitly preserve care-stage boundaries within the timeline, we introduce special boundary tokens, including OUTPATIENT-START, OUTPATIENT-END, EMERGENCY-START, and EMERGENCY-END for OUTP and ED stages, respectively.

Demographic attributes such as gender and race are treated as static events without timestamps and are prepended to the beginning of each sequence. Patient age is computed at the start of each stay and represented as a dedicated event token, AGE-AT-ADMISSION, with an associated numeric value. If a patient dies during or after discharge, a special timestamped token, MEDS_DEATH, is appended to the sequence. We also include special tokens required for language modeling objectives, such as PAD, MASK, CLS, and UNK, where applicable. Table S1 summarizes the different event types present in the patient timeline along with their frequencies.

For events associated with numeric values, such as laboratory tests and ICU chart events, we attach the corresponding numeric measurements. For events without numeric values, including PROCEDURE-ICD and DIAGNOSIS-ICD, we associate a learnable *null* parameter. To encode temporal information, we compute the time difference between consecutive events in minutes and attach this value to each event. Because time gaps may vary substantially both within and across visits, we scale time deltas using the transformation defined in Equation 1, which maps values to the range $[0, 1]$ and prevents extreme magnitudes. Temporal representations are then processed using Time2Vec [71] during training.

$$\Delta t'_i = \frac{\log(1 + \Delta t_i)}{\log(\Delta t_{max})} \quad (1)$$

where Δt_i is the consecutive time difference at event i , and Δ_{max} is the maximum time difference present in the dataset.

To further contextualize each event within the patient timeline, we introduce three additional categorical representations, care stage, visit order, and event type, as illustrated in the bottom rows of Figure S1. The care stage embedding explicitly encodes the clinical context in which each event occurs, distinguishing between outpatient (OUTP), emergency department (ED), inpatient hospitalization (HOSP), ICU stay (ICU), and artificial gap (GAP) events. This representation enables the model to differentiate identical medical events occurring under different clinical settings, which often carry distinct semantic meanings.

	Visit 1											Visit 2									
	OUTP		GAP	ED		HOSP					GAP	ED		HOSP			ICU			HOSP	
Concept	L ₁	L ₂	T	L ₁	L ₂	L ₃	P ₁	P ₂	D ₁	D ₂	T	L ₁	L ₂	P ₁	M ₁	M ₂	P ₂	M ₃	M ₄	D ₁	D ₂
Value	x_l	x_l	null	x_l	x_l	x_l	null	null	null	null	null	x_l	x_l	null	null	null	null	null	null	null	null
Time	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt	Δt
Stage	1	1	0	2	2	3	3	3	3	3	0	2	2	3	3	3	4	4	4	3	3
Visit	1	1	0	1	1	1	1	1	1	1	0	2	2	2	2	2	2	2	2	2	2
Type	1	1	2	1	1	1	3	3	4	4	2	1	1	3	5	5	3	5	5	4	4

Figure S1: Sample patient timeline consisting of two visits. We illustrate the main components constituting the patient timeline in our implementation including medical events, numeric values, time values, care stage representation, visit order representation, and event type representation. Abbreviations, OUTP: Outpatient, ED: Emergency Department, HOSP: Hospital Admission, ICU: Incentive Care Unit, L: Labs, T: TIME-GAP, P: Procedure, M Medication, D: Diagnosis, x_a numeric value. This is a sample timeline created randomly for illustration purposes and does not reflect real data.

In addition, we assign a visit order index to each event, indicating the chronological visit number to which it belongs. The visit order is incremented across care episodes and reset for time gap tokens, allowing the model to reason over longitudinal disease progression across multiple encounters rather than treating the timeline as a flat sequence. Finally, the event type embedding encodes the high-level category of each event, such as laboratory tests, medications, diagnoses, procedures, administrative markers, or special tokens. Below, we provide a formal definition of each component in that define a complete event:

1. **Concept Embedding:** Each clinical concept c_{nk} (e.g., diagnosis, lab test, procedure, medication) is associated with a learnable semantic embedding: $\mathbf{u}_{c_{nk}} \in \mathbb{R}^d$.
2. **Value Embedding:** For events with a numerical measurement v_{nk} , the value is normalized and projected, such that $\mathbf{v}_{nk} = \theta(v_{nk}) \in \mathbb{R}^d$, and $\theta(\cdot)$ is a multi-layer perceptron. Events without values are assigned a learnable (null-value) parameter.
3. **Time Encoding:** Each timestamp t_{nk} is mapped into a temporal embedding representing the local temporal information as time deltas between consecutive events, such that $\mathbf{t}_{nk} \in \mathbb{R}^d$.
4. **Visit Order Embedding:** To capture patient-level longitudinal ordering, each event inherits a visit index embedding: $\mathbf{r}_{\text{visit}(n)} \in \mathbb{R}^d$, where the indices are encoded as continuous learnable positional vectors. This allows the model to distinguish between early and late episodes in the patient trajectory (global temporal representation).
5. **Care Stage Embedding:** Each visit belongs to a clinical stage (e.g., outpatient, emergency, inpatient, ICU). We assign a learned embedding: $\mathbf{s}_{\text{stage}(n)} \in \mathbb{R}^d$ to enable the model to incorporate care-context information, e.g., ICU vs. emergency events carry different clinical semantics.
6. **Type Embedding:** Each event also receives a higher-level event type identifier that encodes the data source from which it originates (ICU chart event, ICU fluid output, etc.): $\mathbf{w}_{\text{type}(nk)} \in \mathbb{R}^d$. This provides local structural information that allows the model to differentiate heterogeneous event types.

By jointly modeling care stage, visit order, and event type, the patient timeline representation preserves structural, temporal, and semantic distinctions across events, facilitating more expressive and context-aware representation learning over heterogeneous EHR sequences.

B.4 Query/History Definition

For each prediction instance in the downstream cohort, we decompose the patient timeline into a *query* segment and a corresponding *history* segment to enable retrieval-based conditioning. The query represents the patient’s current clinical context at the time of prediction and consists of events observed within a task-specific observation window. For details on the observation windows used for each task, we refer the reader to Section C.2 and Table S2. All events occurring strictly before the

Table S1: Summary of the various events present in the patient timeline. We summarize all events present in the patient timeline and list the count along with representative examples. The == sign indicates that event type and example are identical.

Event type	# of unique events	Example
Administrative Events		
OUTPATIENT-START	1	==
OUTPATIENT-END	1	==
EMERGENCY-START	1	==
EMERGENCY-END	1	==
ADMISSION-AT-HOSPITAL	1	==
ADMISSION-AT-ICU	1	==
ADMISSION-LOCATION	12	ADMISSION-LOCATION//TRANSFER FROM HOSPITAL
ADMISSION-TYPE	9	ADMISSION-TYPE//EW EMER
AGE_AT_ADMISSION	1	==
DISCHARGE-FROM-HOSPITAL	1	==
DISCHARGE-FROM-ICU	1	==
DISCHARGE-LOCATION	14	DISCHARGE-LOCATION//HOME
Static Events		
Gender	2	GENDER//F
Race	11	RACE//HISPANIC
MEDS_DEATH	1	==
Medical Events		
LAB	851	LAB//51237//Blood//INR(PT)
MEDICATION	7486	MEDICATION//heparin
MICROBIOLOGY	169	MICROBIOLOGY//90039//URINE CULTURE
PROCEDURE-ICD	13711	PROCEDURE-ICD//3E0G76Z
DIAGNOSIS-ICD	21135	DIAGNOSIS-ICD//K429
DRG	1087	DRG//APR//228
ICU Events		
ICU-CHART	2311	ICU-CHART//220228//Hemoglobin
ICU-FLUID-OUTPUT	71	ICU-FLUID-OUTPUT//226559//Foley
ICU-PROCEDURE	151	ICU-PROCEDURE//225752//Arterial Line
ICU-INFUSION	327	Norepinephrine infusion
Special Tokens		
PAD	1	==
MASK	1	==
CLS	1	==
UNK	1	==
Time Tokens		
Week tokens: W1-W3	3	TIME-GAP//1-W
Month tokens: M1-M12	12	TIME-GAP//11-M
Year token: 1Y-Y+	1	TIME-GAP//1-Y+

query window constitute the patient history and are segmented into fixed-length units that serve as retrievable candidates. Figure S2 illustrates the query/history definitions for the various prediction windows considered in this work. This formulation mirrors real-world clinical reasoning, in which decisions are made based on the present encounter while selectively consulting relevant past visits. By explicitly separating query and history, the model is encouraged to retrieve and integrate only the most informative historical context, rather than uniformly encoding the entire past, enabling scalable reasoning over long and heterogeneous patient trajectories.

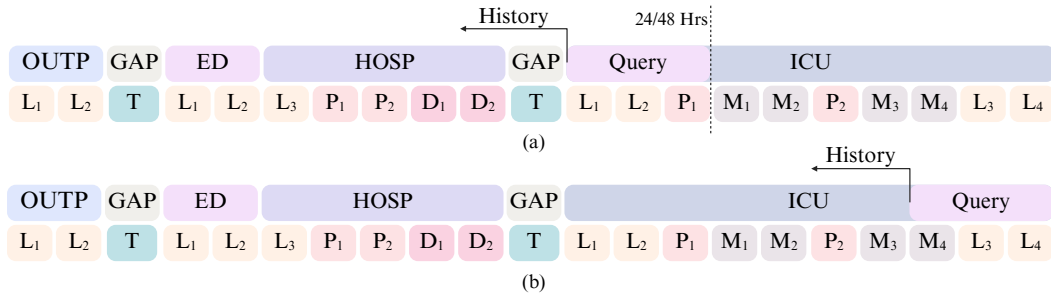


Figure S2: Illustration of query and history definition (a): Query/history boundaries for tasks with time-bounded prediction window. (b) Query/history boundaries task with entire stay prediction window.

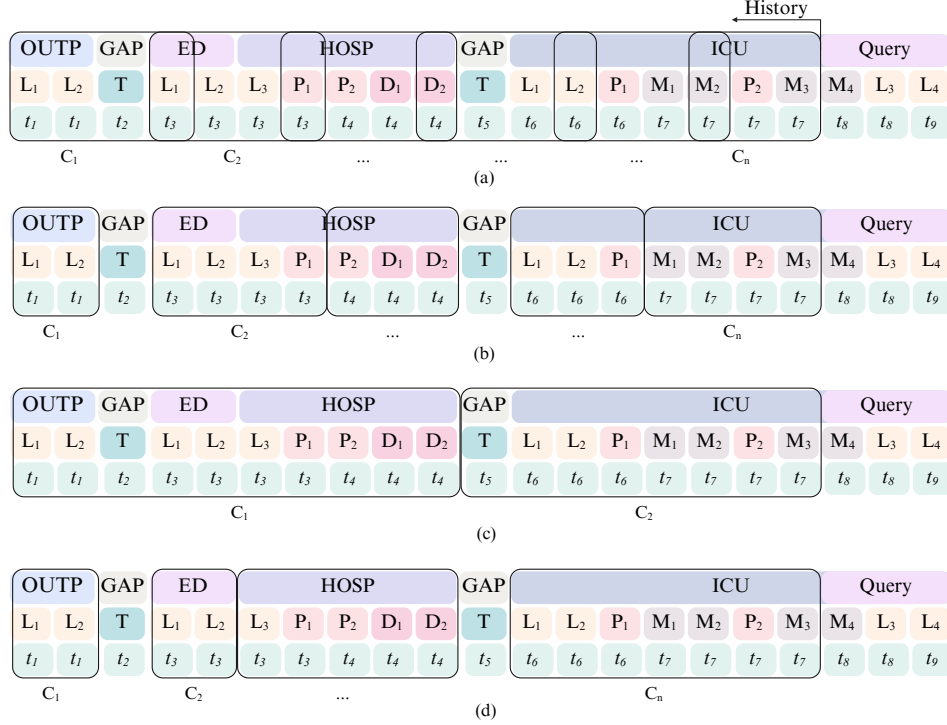


Figure S3: Visual illustration of the chunking strategies employed with EHR-RAGp. (a) Event-based chunking. (b) Time-based chunking. (c) Visit-level chunking. (d) Care-stage chunking.

B.5 Chunking

Structured EHR trajectories are inherently irregular, with clinical events occurring at variable temporal resolutions and across diverse care settings. To organize longitudinal patient history into retrievable units, EHR-RAGp partitions the timeline into clinically coherent chunks. We investigate four chunking strategies: *event-based*, *time-based*, *visit-level*, and *care-stage* chunking. These strategies provide complementary views of patient history by capturing different temporal and clinical granularities. Figure S3 visually illustrates these chunking strategies. Below, we formally outline our proposed chunking strategies:

- **Event-Based:** Events are segmented into fixed-size groups based on event count. Each chunk contains a predefined number of embedded clinical events. Consecutive chunks overlap by a fixed number of events to preserve continuity across chunk boundaries.
- **Time-Based:** Events are segmented into fixed time windows, for example 6-hour, 12-hour, 24-hour blocks. Each chunk contains all events occurring in its temporal window.
- **Visit-level:** Each hospital visit, admission, or encounter is treated as a candidate chunk. This aligns with natural clinical boundaries: emergency department stays, inpatient admissions, and outpatient encounters.
- **Care-Stage:** Events are grouped by clinical stage (outpatient, emergency, inpatient, ICU), ensuring that each chunk reflects a coherent care stage.

In cases where the chunk length is shorter than the context window, we pad the sequence to match the context length. In cases where the chunk length is greater than the context length, we recursively impose *Event-Based Chunking* with proper metadata handling.

C Experimental Setup

C.1 Data Splitting

To ensure a fair evaluation and prevent information leakage across training stages, we adopt a strict data splitting strategy. All data splits are performed at the *patient* level via *subject_id* field rather than the admission or visit level to ensure that no clinical information from the same patient appears in more than one split. For downstream prediction tasks, we partition the eligible ICU cohort into mutually exclusive training, validation, and test sets with proportions of 70%, 10%, and 20%, respectively. The test set is held out entirely and is not used during any stage of model development, including pretraining, hyperparameter tuning, or early stopping. This guarantees that the pretrained representations do not indirectly encode information from evaluation patients. Instead, the validation set is used exclusively for model selection and hyperparameter optimization. Further, for pretraining we hold out 5% as validation set to examine the pretraining quality. This strict separation across patients and training stages ensures that reported performance accurately reflects the model’s ability to generalize to unseen patients and supports reliable comparison across baselines.

C.2 Downstream Tasks

- **In-Hospital Mortality:** This task aims to predict whether a patient will die during the current hospital admission as a binary classification problem, using information available up to a predefined observation window. It is a widely used benchmark for evaluating clinical risk prediction models. In our setup, we condition the in-hospital mortality task on the first 48 hours spent in the ICU, following [25, 26, 9].
- **1-Year Mortality Post-Discharge:** This task focuses on predicting whether a patient will die within one year following hospital discharge or no. It captures longer-term outcomes and reflects the model’s ability to leverage both acute and chronic patterns in a patient’s longitudinal history. We consider the entire ICU stay data as a valid prediction window [72, 73, 74]. For patients with multiple ICU stay and positive label of mortality within 1-year, we only assign the positive label to all stays that are within one year from the mortality date.
- **Long Length of Stay:** This task predicts whether a patient’s hospital stay will exceed a predefined duration threshold. It serves as a proxy for resource utilization and clinical complexity and is commonly used to assess a model’s ability to identify patients at risk of prolonged hospitalization. We set the length-of-stay threshold to 7 days following [6, 8, 54] and define the prediction window as the first 24 hours spent in the ICU, following [75, 76].
- **ICU Readmission:** This task aims to predict whether a patient will be readmitted to the intensive care unit within a specified time window after ICU discharge. It is an important indicator of care quality and patient instability and requires effective modeling of prior clinical trajectories. We set the readmission window to 30 days post-discharge following [77, 8, 78] and use the entire ICU stay as the prediction window, following [8, 79].

Table S2: Overview of the downstream tasks and the query/ history definition for each task

Task	Label type	Prediction Window	Reference point	Query Window	History Chunk
In-hospital mortality	Binary	48 hours since ICU admission	ICU admission	Last L event $\leq$ ICU admission + 48 hours	All events $<$ query start
Long Length of stay-7 Days	Binary	24 hours since ICU admission	ICU admission	Last L event $\leq$ ICU admission + 24 hours	All events $<$ query start
ICU Readmission-30 Days	Binary	Entire ICU stay	ICU discharge	Last L event $\leq$ ICU discharge	All events $<$ query start
1 Year Mortality	Binary	Entire ICU stay	ICU discharge	Last L event $\leq$ ICU discharge	All events $<$ query start

Table S3: Label distribution per data split for downstream prediction tasks.

Task	Train (stay)		Validation (stay)		Test (stay)	
	N Pos (%)	N Neg (%)	N Pos (%)	N Neg (%)	N Pos (%)	Neg (%)
In-hospital mortality	4, 445 (10.415%)	38, 234 (89.585%)	621 (10.126%)	5, 512 (89.874%)	1, 210 (9.787%)	11, 153 (90.213%)
Long Length of stay-7 Days	5, 810 (13.613%)	36, 869 (86.387%)	835 (13.615%)	5, 298 (86.385%)	1, 637 (13.241%)	10, 726 (86.759%)
ICU Readmission-30 Days	1, 558 (3.651%)	41, 121 (96.349%)	230 (3.750%)	5, 903 (96.250%)	480 (3.883%)	11, 883 (96.117%)
1 Year Mortality	5, 741 (13.452%)	36, 938 (86.548%)	842 (13.729%)	5, 291 (86.271%)	1, 654 (13.379%)	10, 709 (86.621%)

C.3 EHR-RAGp Architecture

EHR-RAGp is built on a RoFormer-base [63] encoder configured with 12 transformer layers, 12 attention heads, a hidden size of 768, and rotary positional embeddings applied at each self-attention layer, with a maximum positional embedding length of 1536. The encoder operates over fixed-length event sequences of $C = 1024$ tokens and produces contextualized token representations, from which sequence-level representations are obtained using either *CLS* pooling or mean pooling. We set the chunk overlap to $O = 12.5\%$ of the original chunk length.

Each event token is represented as the sum of multiple embedding components. In addition to the core event embedding, we incorporate categorical embeddings for care stage, visit order, and event type, all projected to the same hidden dimension (768) and learned jointly with the model. Numerical values associated with events are normalized and projected through a small MLP before being fused with the categorical embeddings. In the absence of a numeric value, we use a learnable null embedding of the same dimensionality. Temporal information is encoded using Time2Vec [71], applied to scaled time-delta features and added to the token representation.

For retrieval, the pretrained backbone encoder is used to embed historical timeline chunks, which are stored in a vector database. During training, the query representation is used to retrieve the top- M most similar history chunks based on cosine similarity in the embedding space. Retrieval is non-parametric and does not involve gradient updates during the lookup stage; however, the encoder itself is fine-tuned end-to-end during downstream training.

The prototype module consists of a learnable set of task-specific prototype vectors defined in the same embedding space as the query and history representations. Both query and retrieved history embeddings are projected into the prototype space, where relevance scores are computed via similarity between history embeddings and prototypes, conditioned on the query representation. These scores are used to weight or filter retrieved history chunks, enabling selective integration of long-range context.

The fusion module combines the query representation with the filtered history representations using a shallow transformer with 2 layers and 4 attention heads. The fused representation is then passed to a task-specific prediction head implemented as a lightweight feedforward network. For all downstream tasks, the prediction head outputs a single logit optimized using binary cross-entropy (BCE) loss. All architectural components beyond retrieval are differentiable and optimized jointly during supervised fine-tuning.

C.4 Pretraining Details

We adopt masked language modeling (MLM) objective applied to structured EHR event sequences for EHR-RAGp pretraining. During pretraining, patient timelines are tokenized into fixed-length sequences. We consider the full patient timeline during pretraining and pass it to the encoder backbone as overlapped chunks of 1024 tokens with 128 tokens of overlap. Hence, we produce dense sequences, whereas padding occurs only in the last chunk to fulfill the sequence length. During pretraining, a subset of 15% of the chunk events is randomly selected for masking following the RoBERTa [60] masking strategy. Specifically, masked positions are replaced with a special [MASK] token with probability 80%, substituted with a random event token with probability 10%, or left unchanged with probability 10%. The model is trained to predict the original event identity at masked positions using a cross-entropy loss over the entire vocabulary of 47,377 tokens.

All embedding components, including event embeddings and categorical embeddings, are trained jointly during this stage. No temporal, or numeric value components are used during pretraining; our objective at this stage is solely to learn contextualized representations of structured EHR sequences. Pretraining is conducted on the full pretraining cohort described in Section B.1.

We use *AdamW* [80] optimizer with an initial learning rate specified using cyclical learning rates following [81], a weight decay of 1×10^{-2} , a batch size of 16, and a cosine annealing learning scheduler. Training is performed for 100 epochs over the pretraining corpus. We monitor pretraining quality using accuracy on a held-out validation set, as described in Section C.1. Early stopping is implemented with a patience of three epochs if validation accuracy does not improve. The resulting pretrained backbone is subsequently reused for downstream fine-tuning and retrieval embedding generation. All downstream evaluation experiments are conducted using 4 NVIDIA A100 GPUs.

C.5 Vector Database Setup

To enable retrieval-augmented conditioning, we construct a patient-specific vector database that stores embeddings of historical visit segments derived from the pretrained EHR-RAGp backbone. We use Facebook AI Similarity Search (FAISS²) as the vector index for EHR-RAGp. For each patient in the downstream cohort, the history portion of the timeline (as defined in Section B.4) is segmented into fixed-length chunks, where each chunk corresponds according to the various chunking strategies proposed in our work. Each chunk is independently encoded using the pretrained backbone encoder, and the resulting pooled representation is stored as a dense vector.

We use mean pooling over the sequence representations and normalize the resulting embeddings prior to storage in the index. The vector index is constructed offline before downstream training and remains fixed during supervised fine-tuning. This design avoids repeated recomputation of historical embeddings and ensures scalable retrieval over long patient histories. During retrieval, we perform similarity search based on cosine similarity between the query embedding and historical chunk embeddings. It is worthy to note as well that we do not store chunked data along with their encoded vector, instead; we keep track of the chunk indices in the full chunked patient timeline. Upon retrieval, we perform online chunking during retrieve selected candidates by similarity search via their indices. This approach allow for faster retrieval and reduce the need for maintaining large database that accommodates the chunked data along with the vector index.

C.6 Downstream Training Details

Table S4 summarizes the hyperparameter settings and search spaces used for downstream training of EHR-RAGp. Core backbone parameters are fixed across all experiments, including the RoFormer-base encoder with a hidden dimension of 768, 12 layers, and 12 attention heads.

Retrieval-related hyperparameters including query chunk size, history chunk size, history chunk overlap, number of retrieved chunks, and similarity metric. all retrieval hyperparameters are held constant with values of 1024, 256, 32, 24, *cosine similarity*, respectively.

For prototypes module, we use fixed prototype dimensionality of 768. For temperature, we experiment with temperature pairs (T_q, T_h) , for prototype assignment of $[(0.025, 0.1), (0.05, 0.2), (0.02, 0.08)]$ and Weighing temperature T_s values of $\{0.1, 0.15, 0.2, 0.25\}$, whereas the number of prototypes is set to be optimized with discrete search space in $\{64, 128, 256, 512\}$.

For fusion, we set up a transformer encoder of 2 layers and 4 attention heads, and dropout of 0.1. We set the pooling strategy to be set during hyperparameter optimization with search space of $\{mean, query\}$.

For training hyperparameters, we experiment with Stochastic Gradient Descent (SGD) optimizer, learning rate in $[1e-5, 5e-4]$, Weight decay in $[1e-3, 1e-2]$ and cosine annealing learning rate scheduler. For usage penalty λ_u , we experiment with fixed values in $\{0.004, 0.005, 0.006, 0.007\}$. Batch size is set to be optimized with discrete search space of $\{8, 12, 16, 20\}$ and maximum epochs remain fixed with values of 75.

For hyperparameters optimization, we run 25 trails for each prediction task using Bayesian search. All downstream evaluation experiments were conducted using a single NVIDIA H100 NVL 94GB GPUs. Maximum batch size with can be used with such a GPU in our experiments is 20 query chunks of length 1024 events along with 24 history chunk of length 256, while average epoch time given the max settings is 38 minutes. It is worthy to note that lower capacity GPUs can be also used with reduction in the max settings mentioned earlier.

D Baselines

D.1 Clinical Baselines

We consider **encoder-based** clinical baselines, as they match the structure of EHR-RAGp as an encoder-based model.

²<https://github.com/facebookresearch/faiss>

Table S4: Hyperparameter settings and search spaces used in EHR-RAGp experiments.

Category	Hyperparameter	Fixed	Search Space
Backbone	Encoder architecture	RoFormer-base	–
	Hidden dimension (d)	768	–
	Number of layers	12	–
	Number of attention heads	12	–
	pooling	–	{mean, CLS}
Retrieval	Query Chunk size	1024	–
	History Chunk size	256	–
	History Chunk overlap*	32	–
	Number of retrieved chunks (M)	24	–
	Similarity metric	Cosine	–
Prototypes	Number of prototypes	–	{64, 128, 256, 512}
	Prototype dimension	768	–
	Prototype temperature (T_q, T_h)	–	{(0.025, 0.1), (0.05, 0.2), (0.02, 0.08)}
	Weighting temperature T_w	–	{0.1, 0.15, 0.2, 0.25}
Fusion	Fusion layers	2	–
	Fusion attention heads	4	–
	pooling	–	{mean, query}
	Dropout	0.1	–
Training	Optimizer	SGD	–
	Usage penalty λ_u	–	{0.004, 0.005, 0.006, 0.007}
	Learning rate	–	[1e-5, 5e-4]
	Weight decay	–	[1e-3, 1e-2]
	Batch size	–	{8, 12, 16, 20}
	Max epochs	75	–

* overlap is used only with event-based chunking strategy

- **DescEmb** [58]: is a text-based framework that represents patient history as textual descriptions of medical events, covering the event type, numeric value (if any), along with its unit of measurement. These sequences are then processed via clinical or general-purpose pretrained language models. The main goal of DescEmb is to reduce reliance on predefined medical code vocabularies through a shared linguistic vocabulary that can be utilized across different EHR datasets. We consider two variants of DescEmb in our experiments, including *BERT Fine-Tune* and *CLS Fine-Tune*.
- **GenHPF** [13]: is an extension of DescEmb that goes beyond the basic description of the medical event type, value, and unit to include all features associated with an event in the raw EHR tables, enhancing medical event representation. For instance, a medication event will cover type, name, frequency, dosage, route, etc. It shares the same objective as DescEmb of being agnostic to differing EHR schemas across institutions.
- **Med-BERT** [32]: is a BERT-based foundation model that is pretrained on structured EHR records for disease prediction tasks. It relies solely on structured diagnosis ICD codes as pretraining and evaluation data. The pretraining approach typically follows masked language modeling as proposed in the BERT paper [27].
- **BEHRT** [29]: is another BERT-based foundation model that learns contextual embeddings of the patient’s structured records in a similar fashion to Med-BERT via masked language modeling using diagnosis codes. However, it differs from Med-BERT by incorporating patient age as a temporal signal.
- **Hi-BEHRT** [30]: is a hierarchical extension of BEHRT that aims to handle longer, more comprehensive patient records. It captures short- and long-term patient patterns from hierarchically segmented medical sequences and is pretrained with the self-supervised BYOL framework [82].
- **CEHR-BERT** [31]: is another BERT-based model that incorporates temporal and visit information to improve disease prediction. It represents patient visits as sequences of medical codes, augmented with artificial time tokens, age embeddings, and time embeddings.

D.2 Transformer-based Baselines

- **RoBERTa** [60]: is a transformer-based language model that builds upon BERT by removing the next-sentence prediction objective and adopting dynamic masking and larger pretraining corpora. RoBERTa proves that careful optimization of the pretraining pipelines significantly improves contextual representation learning, which makes it a strong general-purpose encoder baseline.
- **LongFormer** [61]: is a transformer variant designed to handle long input sequences efficiently through a sparse attention mechanism that combines local windowed attention with task-specific global attention. This design reduces the quadratic complexity of standard self-attention, which in turn enables modeling of substantially longer contexts compared to standard transformers.

- **BigBird** [62]: is another long-context transformer architecture that replaces full self-attention with a sparse attention pattern composed of random, local, and global attention. BigBird is theoretically proven to be a universal approximator while enabling scalable processing of long sequences, which making it suitable for tasks that involve extended contextual dependencies.
- **RoFormer** [63]: is a transformer model that introduces rotary positional embeddings (RoPE) to encode relative positional information directly into the self-attention mechanism. This approach improves extrapolation to longer sequences and better captures relative token relationships, which makes it effective for sequence modeling tasks with variable-length inputs.

D.3 Long Context Baselines

- **ModernBERT** [59]: is a re-engineered BERT-style encoder that incorporates recent architectural and training improvements, including optimized attention implementations, improved normalization, and enhanced efficiency on modern hardware. ModernBERT aims to retain the strengths of encoder-based transformers while achieving better performance and scalability.
- **EHRMamba** [10]: typically follows the embedding structure of CEHR-BERT but uses the state-space architecture MAMBA [83] instead of a BERT encoder. The use of a state-space model aims to enable efficient processing of relatively long EHR sequences compared to transformer models.

D.4 Retrieval-based Baselines

- **REMed** [54]: is a retrieval-based foundation model that processes textual descriptions in a similar fashion to both DescEmb and GenHPF. It operates by randomly sampling events from patient history and ranking them to retrieve the most relevant ones for different downstream tasks. In contrast to our proposed model, REMed operates at the medical event level, whereas EHR-RAGp operates at historical segment level to cover broad history beyond limited set of events, 128 in case of REMed.
- **Vanilla EHR-RAGp**: refers to the retrieval-augmented version of the proposed framework without the prototype-guided alignment module. In this setting, historical chunks are retrieved solely based on embedding similarity between the query and patient history, and the retrieved chunks are aggregated directly without prototype-aware refinement. This baseline isolates the contribution of the prototype-guided retrieval mechanism and allows evaluation of how latent prototype alignment improves retrieval quality and downstream prediction performance beyond standard semantic retrieval alone.

D.5 Large Language Model Baselines

- **Qwen2.5-7B-Instruct** [65]: is a general-purpose instruction-tuned large language model designed for conversational reasoning and text understanding across diverse domains.
- **Mistral-7B-Instruct** [66]: is an instruction-following variant of the Mistral family of open-source large language models optimized for efficient inference and general reasoning capabilities.
- **MedGemma-1.5-4B-it** [67] is a medically adapted instruction-tuned language model developed for healthcare and biomedical applications. It is designed to support clinical reasoning and medical text understanding, and is evaluated here as a medical LLM baseline.
- **BioMistral-7B** [68]: is a biomedical domain-adapted variant of Mistral trained on large-scale biomedical corpora to improve performance on medical and scientific language tasks.

D.6 LLM Evaluation Protocol

We evaluated both general-domain and medical-domain large language models using a zero-shot next-token prediction framework for clinical risk prediction. For each downstream task, patient timelines were converted into sequential textual event representations and truncated according to predefined prediction windows. To ensure fair comparison with encoder-based baselines, the same

temporal slicing strategy and prediction horizons were used across all models. A fixed precomputed slices 1024 clinical events along with their numeric values events used.

Each patient sequence was formatted as a chronological list of clinical events and passed directly to the language model without supervised fine-tuning. We used causal language modeling inference and extracted the logits corresponding to the “Yes” and “No” tokens from the next-token distribution. All other token logits were ignored, and probabilities were normalized over the binary “Yes/No” subset using a softmax operation. The normalized probability assigned to the “Yes” token was used as the prediction score. We present the prompt used for each task in The prompts used for zero-shot evaluation are shown in Box D.6.

Performance was evaluated using AUROC and AUPRC. Confidence intervals were estimated using nonparametric bootstrap resampling with 1000 iterations.

LLM Prompt Templates

In-Hospital Mortality

You are an expert clinical risk prediction model using electronic health records.

--- PATIENT DATA ---

Electronic Health Records:
{ehr_text}

--- TASK ---

Will this patient die during this hospital admission?

Answer only using one word: Yes or No.

Answer:

1-Year Mortality

You are an expert clinical risk prediction model using electronic health records.

--- PATIENT DATA ---

Electronic Health Records:
{ehr_text}

--- TASK ---

Will this patient die within 1 year after hospital discharge?

Answer only using one word: Yes or No.

Answer:

ICU Readmission within 30 Days

You are an expert clinical risk prediction model using electronic health records.

--- PATIENT DATA ---

Electronic Health Records:
{ehr_text}

--- TASK ---

Will this patient be readmitted to the ICU within 30 days after ICU discharge?

Answer only using one word: Yes or No.

Answer:

Long Length of Stay > 7 Days

You are an expert clinical risk prediction model using electronic health records.

--- PATIENT DATA ---

Electronic Health Records:
{ehr_text}

--- TASK ---

Will this patient have a hospital length of stay longer than 7 days?

Answer only using one word: Yes or No.

Answer:

Table S5: Performance comparison between EHR-RAGp and LLM baselines

Model	ICU-Readmit 30d		In-hospital Mortality		Long LOS 7d		1YR Mortality	
	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)	AUROC (CI)	AUPRC (CI)
General LLMs								
Qwen2.5-7B-Instruct	0.533 (0.508, 0.557)	0.041 (0.036, 0.045)	0.747 (0.733, 0.761)	0.367 (0.342, 0.391)	0.537 (0.522, 0.551)	0.147 (0.138, 0.157)	0.548 (0.534, 0.561)	0.142 (0.134, 0.150)
Mistral-7B-Instruct	0.515 (0.490, 0.540)	0.039 (0.035, 0.045)	0.723 (0.708, 0.740)	0.313 (0.287, 0.340)	0.610 (0.595, 0.624)	0.186 (0.174, 0.200)	0.511 (0.498, 0.526)	0.130 (0.124, 0.138)
Clinical LLMs								
medgemma-1.5-4b-it	0.500 (0.472, 0.525)	0.039 (0.035, 0.044)	0.594 (0.578, 0.612)	0.138 (0.127, 0.151)	0.618 (0.604, 0.632)	0.191 (0.179, 0.206)	0.531 (0.517, 0.547)	0.146 (0.137, 0.155)
BioMistral-7B	0.509 (0.479, 0.535)	0.041 (0.036, 0.047)	0.594 (0.578, 0.610)	0.147 (0.134, 0.161)	0.523 (0.509, 0.538)	0.136 (0.129, 0.145)	0.516 (0.502, 0.530)	0.136 (0.128, 0.144)
EHR-RAGp (Ours)	0.747 (0.724, 0.768)	0.156 (0.128, 0.189)	0.940 (0.933, 0.945)	0.716 (0.693, 0.738)	0.885 (0.876, 0.893)	0.628 (0.603, 0.652)	0.821 (0.811, 0.831)	0.396 (0.372, 0.422)

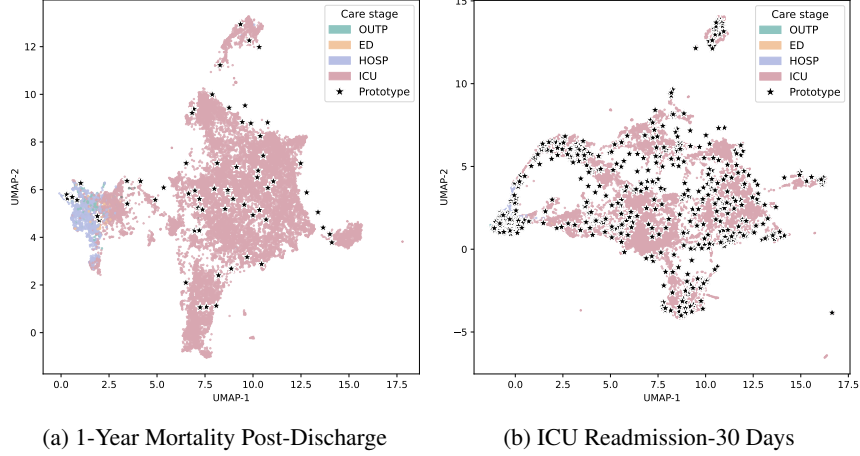


Figure S4: UMAP visualization of history chunk embeddings and prototypes for (a) long length of stay and (b) in-hospital mortality tasks.

E Additional Results

E.1 LLM Baseline Results

Table S5 compares EHR-RAGp against both general-purpose and medically adapted LLMs evaluated in a zero-shot setting. Across all tasks, LLM baselines substantially underperform specialized EHR models, with particularly weak performance on ICU readmission and long-term mortality prediction. For example, the strongest LLM baseline, Qwen2.5-7B-Instruct, achieves an AUPRC of only 0.041 for ICU readmission compared to 0.156 for EHR-RAGp. Similarly, medical LLMs such as MedGemma and BioMistral do not demonstrate meaningful improvements over general LLMs, despite domain adaptation. Overall, these results suggest that current zero-shot LLMs struggle to effectively reason over structured longitudinal EHR trajectories, highlighting the importance of architectures specifically designed for temporal clinical data and patient-history retrieval.

E.2 Latent Representation Structure

Figure S4 presents UMAP projections of history chunk embeddings for 1-Year Mortality Post-Discharge and ICU Readmission-30 Days.

E.3 Qualitative Analysis of Prototype-Guided Retrieval

This section provides a qualitative analysis of the prototype-guided retrieval mechanism in EHR-RAGp. To illustrate the behavior of the model, we present two representative case studies from the in-hospital mortality task, corresponding to positive and negative prediction examples. For each case, we analyze prototype usage, query-history alignment, chunk-level relevance weighting, and the contribution of visits and care stages to the final prediction. These examples provide insight into how EHR-RAGp organizes longitudinal patient history and selectively integrates clinically relevant context during inference.

E.3.1 Case of Positive Example

Figure S5 presents a representative positive example from the in-hospital mortality task, corresponding to a correctly predicted positive case ($\text{label}=1$, $\text{pred}=1$) with high confidence ($\text{prob}=0.998$) for a query segment originating from Visit V2 in the patient timeline. Panel (a) shows the query prototype distribution across the top activated prototypes, where prototype $P17$ receives the highest assignment probability. Panel (b) illustrates the alignment between retrieved history chunks and the top query prototypes, revealing consistent prototype agreement across several retrieved chunks, particularly among the early ICU-related chunks. Panel (c) shows the final prototype-guided chunk weights, where the model assigns the highest importance to chunks originating from the second visit and predominantly associated with the ICU care stage. Chunks from lower-acuity stages such as ED and outpatient care receive comparatively lower weights. Panels (d) and (e) further aggregate these contributions at the visit and care-stage levels, showing that the prediction is primarily driven by the recent visit trajectory and high-acuity clinical context. Overall, the example illustrates how EHR-RAGp selectively emphasizes clinically severe and temporally relevant historical context when forming the final prediction.

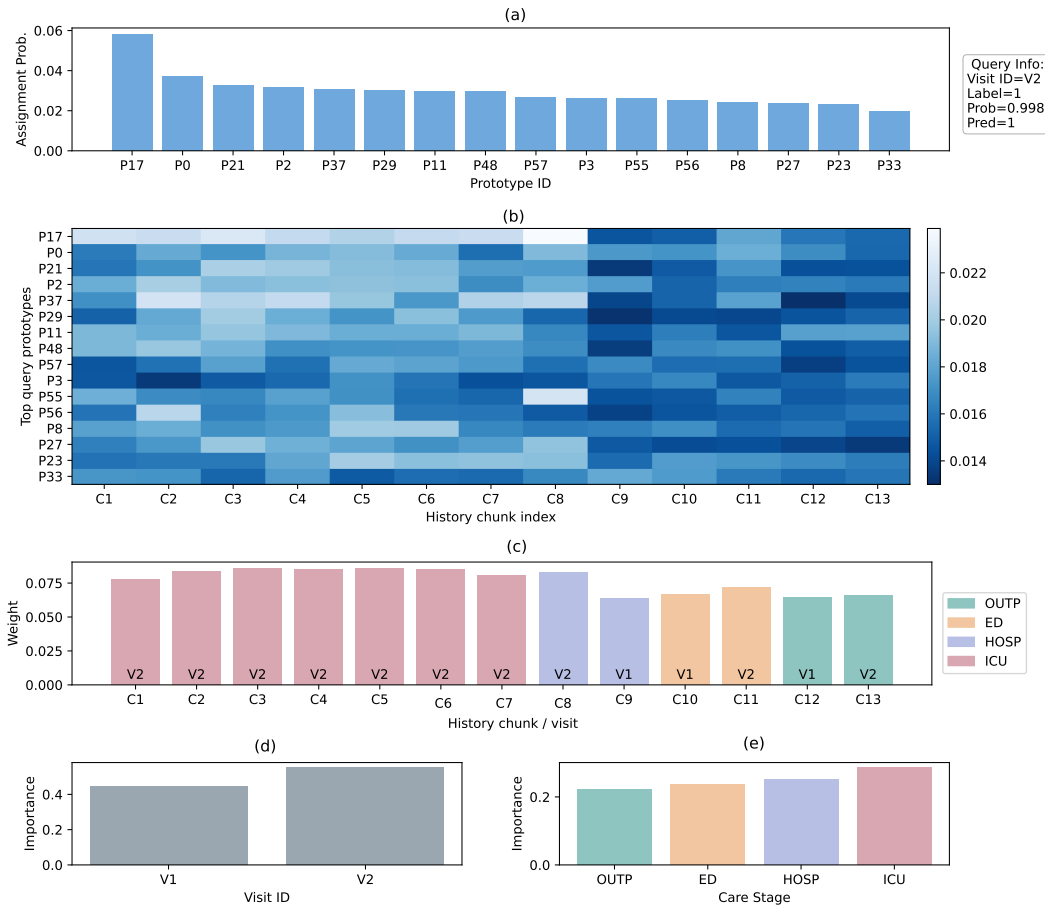


Figure S5: Qualitative analysis of prototype-guided retrieval for an in-hospital mortality positive example. (a) Top prototype assignments for the query segment. (b) Alignment between retrieved history chunks and the query prototypes. (c) Prototype-guided relevance weights assigned to retrieved chunks. (d) Aggregated contribution of retrieved visits. (e) Aggregated importance across care stages.

E.3.2 Case of Negative Example

Figure S6 presents a representative negative example from the in-hospital mortality task, corresponding to a correctly predicted negative case ($\text{label}=0$, $\text{pred}=0$) with low predicted probability ($\text{prob}=0.006$) for a query segment originating from Visit V5 in the patient timeline. Panel (a) shows

the query prototype distribution across the top activated prototypes, with a more diffuse assignment pattern, indicating less concentration around highly activated latent modes. Panel (b) illustrates the prototype agreement between the query and retrieved history chunks, where alignment is more uniformly distributed across chunks without a dominant subset receiving consistently strong activations. In panel (c), the final prototype-guided chunk weights are relatively balanced across visits and care stages, with no strong emphasis on ICU-related segments. Panels (d) and (e) further confirm this behavior, showing that importance is distributed across multiple visits and care stages rather than concentrated around a single high-acuity trajectory. Overall, this example illustrates how EHR-RAGp leverages broader longitudinal history across multiple visits and care stages to support a confident negative prediction, without relying on a single dominant high-acuity trajectory.

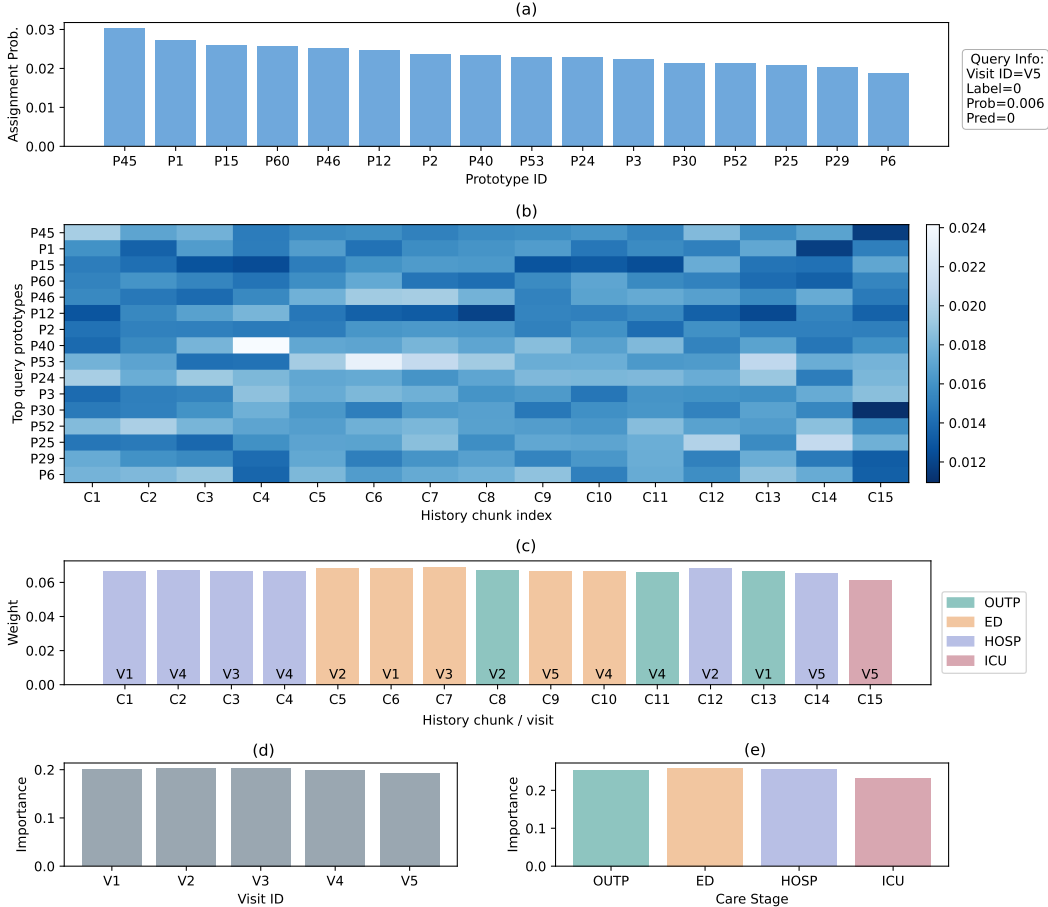


Figure S6: Qualitative analysis of prototype-guided retrieval for an in-hospital mortality negative example. (a) Top prototype assignments for the query segment. (b) Alignment between retrieved history chunks and the query prototypes. (c) Prototype-guided relevance weights assigned to retrieved chunks. (d) Aggregated contribution of retrieved visits. (e) Aggregated importance across care stages.

E.4 Analysis of Prototype Regularization

This section analyzes the effect of prototype regularization on the behavior of EHR-RAGp. Specifically, we compare the model with and without regularization from two perspectives: (1) global prototype utilization patterns, illustrating how regularization affects the distribution and diversity of prototype assignments, and (2) qualitative case studies, showing how regularization influences prototype-guided retrieval and chunk weighting for individual patient trajectories. Together, these analyses highlight the role of regularization in preventing prototype collapse and encouraging more balanced use of the latent prototype space.

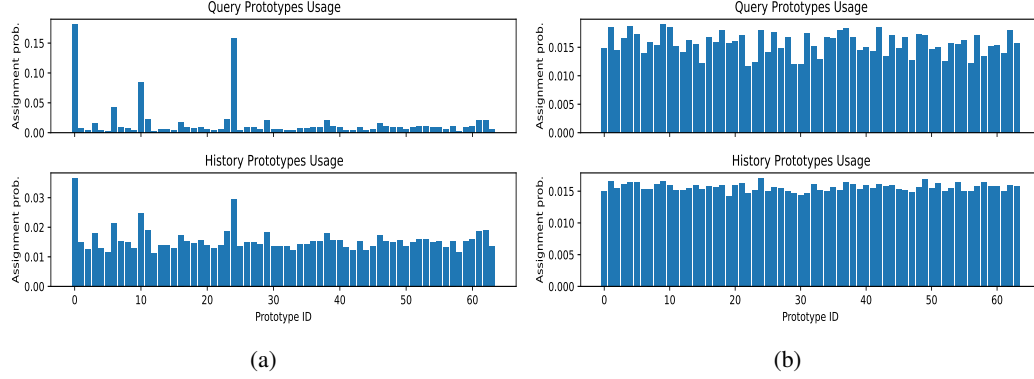


Figure S7: Global prototype utilization patterns. (a) Without Regularization. (b) With regularization.

E.4.1 Global Prototypes Usage

Figure S7 illustrates the effect of prototype regularization on global prototype utilization. Without regularization as in Figure S7a, query assignments exhibit highly imbalanced usage patterns, where a small subset of prototypes receive disproportionately large assignment probabilities while many prototypes remain minimally utilized. A similar, although less severe, imbalance is also observed for history chunk assignments. This behavior indicates partial prototype collapse, where the model concentrates representation learning around a limited number of latent prototypes. In turn, such a situation makes agreement between history and query rather difficult due to the sharper distribution of the query as compared to the history assignments. In contrast, enabling regularization, as in Figure S7b, produces a substantially more uniform assignment distribution across prototypes for both queries and history chunks. The resulting increase in prototype diversity suggests that regularization encourages broader utilization of the latent prototype space, leading to more balanced representation learning and improved retrieval behavior.

E.4.2 Qualitative Analysis of Prototypes Usage

To further understand the role of usage regularization λ_u , we examine the model behavior with regularization, as in in Figure S8, and without regularization, as in Figure S9. In the settings where usage regularization is activated, prototype assignments are distributed across multiple latent prototypes, resulting in relatively balanced alignment scores and chunk weights across retrieved history segments. This behavior indicates that the model integrates information from several relevant historical chunks rather than relying on a single dominant prototype or retrieval path. On the other hand, without usage regularization, the query representation collapses strongly toward a single prototype (P0), which leads to highly concentrated retrieval behavior where few retrieved chunk receives the majority of the final weight while the remaining chunks contribute minimally. With respect to the model performance, it can be observed that despite the collapse the model can still provide correct prediction but less confident. Overall, these examples qualitatively demonstrate how prototype regularization stabilizes retrieval dynamics, encourages broader utilization of patient history, and prevents highly concentrated dependence on a narrow subset of latent prototypes, which in turn deteriorates the retrieval quality.

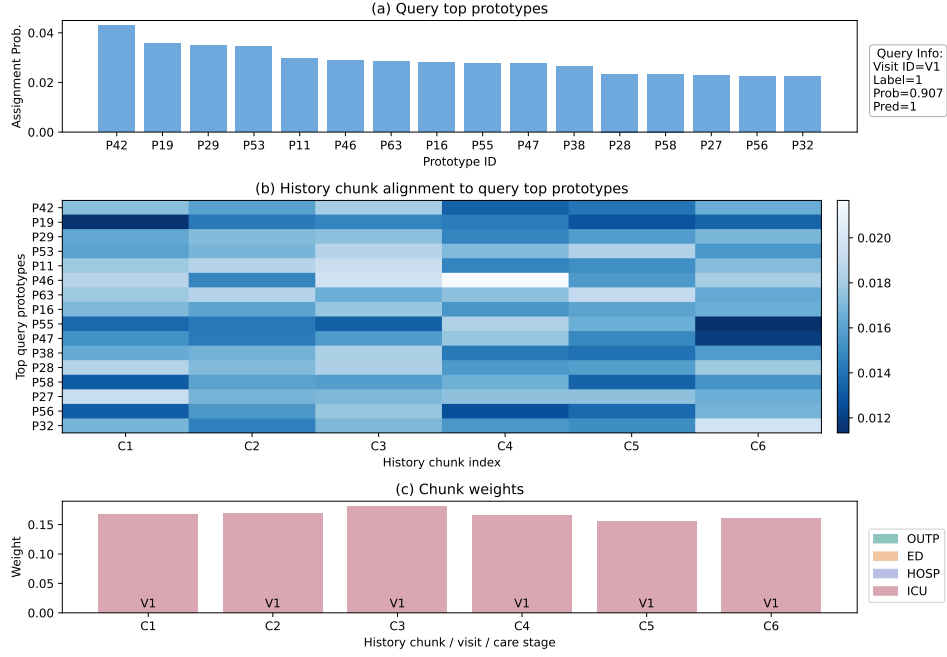


Figure S8: Qualitative analysis of sample prediction for long length of stay (7 days) task with regularization (a) Top prototype assignments for the query segment. (b) Alignment between retrieved history chunks and the query prototypes. (c) Prototype-guided relevance weights assigned to retrieved chunks.

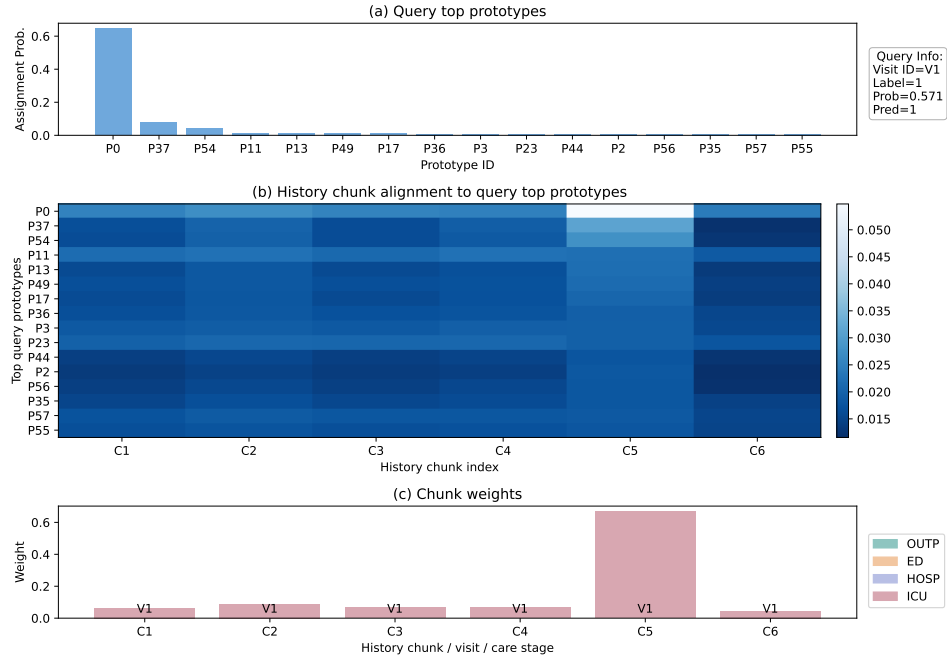


Figure S9: Qualitative analysis of sample prediction for long length of stay (7 days) task without regularization (a) Top prototype assignments for the query segment. (b) Alignment between retrieved history chunks and the query prototypes. (c) Prototype-guided relevance weights assigned to retrieved chunks.